

Module 04 – Multiperiod Modeling

EDA

- Make a nicely formatted table with the needed data on each investment

Investment	Inflow	Outflow	Amount	Return	1	2	3	4	5	6	7	8	9	10
A	1	2	\$	-	2.0%	-1	1							
B	1	3	\$	885.18	4.2%	-1	<-->	1						
E	1	6	\$	225.33	11.0%	-1	<-->	<-->	<-->	1				
A	2	3	\$	-	2.0%		-1	1						
C	2	5	\$	-	6.5%		-1	<-->	<-->	1				
A	3	4	\$	-	2.0%			-1	1					
B	3	5	\$	239.88	4.2%		-1	<-->	1					
D	3	7	\$	432.66	8.7%			-1	<-->	<-->	1			
A	4	5	\$	-	2.0%				-1	1				
A	5	6	\$	-	2.0%					-1	1			
C	5	8	\$	-	6.5%				-1	<-->	<-->	1		
B	5	7	\$	-	4.2%					-1	1			
A	6	7	\$	-	2.0%						-1	1		
E	6	11	\$	-	11.0%					-1	<-->	<-->	<-->	<-->
A	7	8	\$	-	2.0%						-1	1		
B	7	9	\$	470.35	4.2%						-1	<-->	1	
D	7	11	\$	-	8.7%						-1	<-->	<-->	<-->
A	8	9	\$	-	2.0%							-1	1	
C	8	11	\$	-	6.5%							-1	<-->	<-->
A	9	10	\$	490.20	2.0%								-1	1
B	9	11	\$	-	4.2%								-1	<-->
Total Invested in Month 1 ->				\$ 1,110.51										
				Surplus Funds	\$-	\$250.00	\$-	\$250.00	\$250.00	\$-	\$-	\$-	\$-	\$500.00
				Req'd Payments	\$0	\$250	\$0	\$0	\$250	\$0	\$0	\$0	\$0	\$500

Model Formulation

Decision Variables: Amount Invested (x_i in investment i , where $i = a,b,c,d,e$)

Investment	Inflow	Outflow	Amount
A	1	2	\$ -
B	1	3	\$ 885.18
E	1	6	\$ 225.33
A	2	3	\$ -
C	2	5	\$ -
A	3	4	\$ -
B	3	5	\$ 239.88
D	3	7	\$ 432.66
A	4	5	\$ -
A	5	6	\$ -
C	5	8	\$ -
B	5	7	\$ -
A	6	7	\$ -
E	6	11	\$ -
A	7	8	\$ -
B	7	9	\$ 470.35
D	7	11	\$ -
A	8	9	\$ -
C	8	11	\$ -
A	9	10	\$ 490.20
B	9	11	\$ -
Total Invested in Month 1 ->			\$ 1,110.51

Return Rate of Investment

Investment	Months to Mature	Return
A	1	2%
B	2	4%
C	3	6%
D	4	9%
E	5	11%

Time period for each investment:

investment_name	month_can_start_investing
Chocolate Chip Capital	1
Peppermint Profit Partners	1
RockCandy Returns	2
SourPatch Portfolio Group	3
Toffee Turnaround Trust	1

Objective Function: Minimize total out of pocket costs (amount invested in month 1)
= SUM(amount a through E)

Constraints: Amount >= 0
Surplus Funds = Required Payments

Model Optimized for Least Cost out of Pocket

Investment Inflow Outflow	Amount	Return	1	2	3	4	5	6	7	8	9	10
A 1 2	\$	2.0%	-1	1								
B 1 3	\$ 885.18	4.2%	-1	<-->	1							
E 1 6	\$ 225.33	11.0%	-1	<-->	<-->	<-->	<-->	1				
A 2 3	\$	2.0%		-1	1							
C 2 5	\$	6.5%		-1	<-->	<-->	1					
A 3 4	\$	2.0%			-1	1						
B 3 5	\$ 239.88	4.2%		-1	<-->	1						
D 3 7	\$ 432.66	8.7%			-1	<-->	<-->	1				
A 4 5	\$	2.0%				-1	1					
A 5 6	\$	2.0%					-1	1				
C 5 8	\$	6.5%					-1	<-->	<-->	1		
B 5 7	\$	4.2%						-1	<-->	1		
A 6 7	\$	2.0%						-1	1			
E 6 11	\$	11.0%						-1	<-->	<-->	<-->	<-->
A 7 8	\$	2.0%							-1	1		
B 7 9	\$ 470.35	4.2%							-1	<-->	1	
D 7 11	\$	8.7%							-1	<-->	<-->	<-->
A 8 9	\$	2.0%								-1	1	
C 8 11	\$	6.5%								-1	<-->	<-->
A 9 10	\$ 490.20	2.0%									-1	1
B 9 11	\$	4.2%									-1	<-->
Total Invested in Month 1 ->	\$ 1,110.51											
Surplus Funds	\$		\$ -	\$250.00	\$ -	\$250.00	\$250.00	\$250.00	\$ -	\$ -	\$ -	\$500.00
Req'd Payments	\$		\$0	\$250	\$0	\$0	\$0	\$250	\$0	\$0	\$0	\$500

The final model shows a structured investment of inflows and outflows in a way that minimizes additional out-of-pocket expenses by prioritizing reinvestments of surplus funds and minimizing cash flow by using existing investments.

Investment Inflow Outflow	Amount	Return	1	2	3	4	5	6	7	8	9	10
A 1 2	\$	2.0%	-1	1								
B 1 3	\$ 885.18	4.2%	-1	<-->	1							
E 1 6	\$ 225.33	11.0%	-1	<-->	<-->	<-->	<-->	1				
A 2 3	\$	2.0%		-1	1							
C 2 5	\$	6.5%		-1	<-->	<-->	1					
A 3 4	\$	2.0%			-1	1						
B 3 5	\$ 239.88	4.2%			-1	<-->	1					
D 3 7	\$ 432.66	8.7%			-1	<-->	<-->	1				
A 4 5	\$	2.0%				-1	1					
A 5 6	\$	2.0%					-1	1				
C 5 8	\$	6.5%					-1	<-->	<-->	1		
B 5 7	\$	4.2%						-1	<-->	1		
A 6 7	\$	2.0%						-1	1			
E 6 11	\$	11.0%						-1	<-->	<-->	<-->	<-->
A 7 8	\$	2.0%							-1	1		
B 7 9	\$ 470.35	4.2%							-1	<-->	1	
D 7 11	\$	8.7%							-1	<-->	<-->	<-->
A 8 9	\$	2.0%								-1	1	
C 8 11	\$	6.5%								-1	<-->	<-->
A 9 10	\$ 490.20	2.0%									-1	1
B 9 11	\$	4.2%									-1	<-->
Total Invested in Month 1 ->	\$ 1,110.51											
Surplus Funds	\$		\$ -	\$250.00	\$ -	\$250.00	\$250.00	\$ -	\$ -	\$ -	\$ -	\$500.00
Req'd Payments	\$		\$0	\$250	\$0	\$0	\$0	\$250	\$0	\$0	\$0	\$500

Could Have Savings Out of Pocket

\$1,000.00 \$ 110.51 \$ 1,110.51

Investment Summary



The investment summary compares the out-of-pocket savings to the potential savings. Out of pocket savings = \$1,110.51, and the potential savings = \$1,000. In this case, out of pocket savings surplus the potential savings.